

2.3.2 Phase I

In the last section we assumed that we can start with some known feasible basis for the LP. If the LP is given in canonical form (1.5) with nonnegative right hand side b such a basis is easy to obtain by transforming the problem to standard form and using the slack variables s as initial basis ($s = b, x = 0$ is always a feasible solution in this case). For LPs in standard form there are various ways to obtain a starting basis. This procedure is conventionally called *Phase I* or *crash phase* of the simplex algorithm.

An obvious (albeit inefficient) way to handle LPs in standard form is to transform each equation into two inequalities, thereby obtaining an LP in canonical form, and then proceed as above.

A better method is to introduce an artificial variable s_i with coefficients δ_i in each row, setting $\delta_i = \text{sign}(b_i)$, and solve the auxiliary LP

$$\begin{array}{ll} \min & \sum_i s_i \\ \text{s. t.} & Ax + \text{Diag}(\delta)s = b, \\ & x, s \geq 0, \end{array} \quad (2.5)$$

where $\text{Diag}(\delta)$ denotes the diagonal matrix with entries δ_i on the diagonal. Using the basis composed of the artificial variables s we can start the simplex algorithm. If the auxiliary problem has an optimal objective value of 0 we have found a feasible solution for the original problem. From this one can – possibly using some extra pivots and possibly handling rank deficiency of the initial matrix – remove all s_i -variables from the basis to obtain a feasible starting basis for the original problem³.

2.3.3 Detecting Infeasibility

The auxiliary LP (2.5) also provides a simple way to verify whether a given LP in standard form is feasible: If $s = 0$ in the optimal solution then the original LP is feasible.

³Since all s_i remaining in the optimal basis of the auxiliary problem have value 0, such a tableau must be degenerate (see next section) and we can pivot even on negative entries of the reduced non-basis. If after this any artificial variables remain basic, the subsystem of rows indexed by basic artificial variables is trivially satisfied for any choice of the non-basic variables, and hence the whole subset of rows can be deleted.

Algorithm 3 Simplex Phase I

Require: LP in general form:

$$\begin{aligned}
& \max \quad c^\top x \\
& \text{s. t.} \quad Ax = b \\
& \quad \quad Dx \leq d \\
& \quad \quad Fx \geq f \\
& \quad \quad x \geq 0
\end{aligned}$$

Ensure: INFEASIBLE or a feasible simplex basis and a subset of rows of the LP which is equivalent

1: Set up the auxiliary problem

$$\begin{aligned}
z = \min \quad & 1^\top s_1 \quad \quad \quad + 1^\top s_2 \quad \quad \quad + 1^\top s_3 \\
\text{s. t.} \quad & Ax \quad + \text{Diag}(\delta(b))s_1 \quad \quad \quad = b \\
& Dx \quad \quad \quad + \text{Diag}(\delta(d))s_2 \quad \quad \quad \leq d \\
& Fx \quad \quad \quad - \text{Diag}(\delta(f))s_3 \geq f \\
& x, \quad \quad \quad s_1, \quad \quad \quad s_2, \quad \quad \quad s_3 \geq 0
\end{aligned}$$

where $\delta_i = \text{sign}(b_i)$, 1 is the all-one vector and I is the identity matrix in appropriate dimension.2: Solve the auxiliary problem using the simplex method with starting basis (s_1, s_2, s_3) 3: **if** $z > 0$ **then**4: **return** INFEASIBLE5: **else**6: **if** any variables $(s_i)_j$ remain in the basis **then**

7: perform degenerate pivoting to move them to the non-basis

8: **end if**9: **if** any variables $(s_i)_j$ remain in the basis **then**10: delete rows corresponding to basic variables $(s_i)_j$ 11: **end if**12: **return** optimal basis of the auxiliary problem and current set of rows13: **end if**

2.3.4 Termination, Degeneracy and Cycling

Definition 2.4. A feasible tableau is called *degenerate* if the reduced right hand side $\bar{b}$ contains any zero elements. Otherwise it is called *non-degenerate*.

There are two reasons for degeneracy: geometric degeneracy and formulation degeneracy. For the former case consider the top vertex of a square pyramid in dimension 3, incident to 4 hyperplanes. In this case there are multiple bases representing this same point, and the pivots between them have ratio $\lambda_0 = 0$. Geometric degeneracy cannot be avoided. Similarly one may find degeneracy if redundant rows are in the LP formulation, which is often the case for practical models. Such redundancy can in principle be avoided by eliminating the useless rows, but deciding whether a row is redundant requires the solution of an LP.

Degeneracy is an important phenomenon, since it can be the cause for *cycling*, i.e. a succession of pivot operations that return to the same tableau after a number of (necessarily degenerate) steps. Beale [1955] was the first to notice that this can occur for reasonable (and deterministic) pivoting rules.

Example 2.5. Consider the revised tableau (variable names have been suppressed)

10	-57	-9	-24		0
-0.5	5.5	2.5	-9		0
-0.5	1.5	0.5	-1		0
-1	0	0	0		1

and use the steepest edge column pivot rule. If two or more rows qualify in the ratio test, choose the one with smallest row index.

After 6 pivots we arrive at the same tableau as above (except for possible reordering).

Cycling can be avoided by perturbing the problem, or by choosing a specialized pivot rule. We will not discuss perturbation methods here, although they are what is used in practical implementations. Instead we cite

Theorem 2.6 (Bland's (first) rule (see Bland [1977])). *Using the smallest variable index pivot for both column and row pivot selection ensures that the simplex algorithm will not cycle.*

For a proof see Chvátal [1983].